TEACHER GUIDE

ANSWERS AND MARKS

GRADE 11 · INFORMATION AND COMMUNICATION TECHNOLOGY

UNIT 6

Information and Communication Technology and Society



! FOR THE TEACHER ONLY

This booklet has the answers to every exercise in the Unit 6 Student Workbook, in the same order. The exercise letters match the workbook.

i WHAT IS IN THIS GUIDE

- **Marks and pacing** - every exercise and its marks
- **Class marks record** - one page to copy for each class
- **Practicals at a glance** - Practicals 1 - 2, materials, textbook links, a quiz
- **Writing spaces** - what students write in every writing space
- **Answers** - chapters 1 to 4, in the same order as the workbook
- **Unit 6 Test** - the marking scheme, common mistakes, textbook notes

★ BEFORE THE FIRST LESSON

- ☐ A computer with the Internet and a projector, and an updated virus guard.
- ☐ The school computer lab: its UPS, surge protector, locks and CCTV to show.
- ☐ The notes of the writing spaces (the table in this guide) for each lesson.

"Teach your students to use ICT wisely, safely and honestly."

- THE KING OF ICT

TEACHER GUIDE

Marks and Pacing

You decide how much class time each part needs

Chapter	Exercise	Marks
Chapter 1	A Match them	6
	B Fill in the blanks	8
	C True or false	6
	D Choose the correct answer	5
Chapter 2	A Which legal issue?	8
	B Citing, quoting or referencing?	6
	C Choose the correct answer	5
	D Answer these	4
Chapter 3	A Physical or logical?	8
	B Match the harmful software	8
	C True or false	6
	D Choose the correct answer	5
	E Word search	10
Chapter 4	A Which health issue?	8
	B Where does it hurt?	4
	C Choose the correct answer	5
End	Unit 6 Test	40
TOTAL		**142**

! SUGGESTED SHAPE OF ONE CHAPTER

READ	Teach the chapter pages. Students read with you and underline the key words.
DO	The practical tasks on real computers, in pairs.
WRITE	The written exercises. Walk around and help.
MARK	Mark the work together. Fill the tick boxes.

★ TEACHING TIPS

- Start Chapter 1 with a question: who has watched a 3D film? What did the 3D spectacles do?
- Show a short part of a cartoon film. Ask why adults and children both enjoy it (the 3D feature, new software).
- If the lab has a music program, show recording, rhythm and mixing for two minutes (digital audio materials).
- Ask where pupils have seen a driving or flight simulator: simulation games are used for training.
- Before Practical 1, check that the lab computers have an updated virus guard.

TEACHER GUIDE

Class Marks Record

Copy this page for each class

No	Student name	Ch 1 (25)	Ch 2 (23)	Ch 3 (37)	Ch 4 (17)	Test (40)	Total (142)
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
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25							
26							
27							
28							
29							

TEACHER GUIDE

Practicals at a Glance

The two activities of textbook pages 178 and 179

Practical (activity)	Students do	Check that ...
1 (page 178)	search the Internet for virus guards that are not in the list of 6.10; tabulate the manufacturer and the date of manufacture	pupils use secure websites and download or install nothing; the names and the manufacturers are written correctly.
2 (page 179)	open the three websites of the textbook and list the computer related services of each one	each URL is checked before it is used and no personal information is entered on any website.

SAFETY AND MATERIALS

- The textbook has two activities in chapter 6 (pages 178 and 179); Practicals 1 and 2 follow them.
- Computers with the Internet and an updated virus guard. Nothing is downloaded or installed.
- Check beforehand that http://www.gov.lk/web/, http://www.engage.icta.lk and https://www.techcert.lk/si/ open in the lab. If one has moved, use the organization's present website.
- Each writing space takes about five minutes: read the note out or write it on the board.

Textbook links

Workbook	Textbook section	Textbook pages	Activities
Chapter 1 - ICT in entertainment (6.1 - 6.2)	6.1	168 - 170	-
Chapter 2 - Ethical and legal issues (6.3 - 6.5)	6.2, 6.2.1, 6.2.2	170 - 172	-
Chapter 3 - Protecting ICT infrastructure (6.6 - 6.11)	6.2.3, 6.2.4	172 - 179	178, 179
Chapter 4 - Health issues and ergonomics (6.12 - 6.13)	6.2.5	179 - 180	-
Unit 6 in One Page	-	168 - 180	-

The textbook numbers its sections 6.1 (ICT in entertainment) and 6.2 - 6.2.5 (problems associated with the use of ICT). The workbook numbers its own sections 6.1 - 6.13 on through the unit.

TEACHING TIPS

- Draw Figure 6.8 on the board and let pupils give one example for each of the seven issues.
- Use a lost mobile phone as the example of data theft: which personal information (PII) is on it?

QUICK ORAL QUIZ - START OR END A LESSON WITH IT

Ask the class ...	Answer
1. What do you need to view a 3D film?	**3D spectacles**
2. Name two uses of simulation games.	**Military training, sports training (any two)**
3. Name three issues of Figure 6.8.	**Legal, values, physical ... (any three)**
4. What is PII?	**Personally Identifiable Information**
5. What is a patent?	**A license issued by the state for innovations**
6. What is plagiarism?	**Displaying another's creative work as one's own**
7. What does a UPS do in a power failure?	**Its battery supplies power**
8. When does a surge protector act?	**When the supply exceeds 240 V**
9. Which harmful software does not spread by itself?	**The Trojan horse**
10. What do "ergon" and "nomos" mean?	**Job or action; law**

TEACHING TIPS

- Never let pupils share passwords. Show how a stolen password leads to unauthorized access.
- Explain patents with a new product the pupils know: who owns it, and who gives the license?
- Give a short paragraph from a website. Pupils quote one sentence, cite the owner and write the reference.
- Praise projects written in the pupils' own words with a reference list. That is the opposite of plagiarism.
- Walk round the computer lab: find the UPS, the surge protector, the locked doors and the CCTV camera.
- Physical security protects the devices; logical security protects the data. Write the two lists side by side.
- Make a strong password together on the board: letters, symbols and numbers mixed (textbook page 174).
- Open the firewall settings of a lab computer and show that the software firewall is on (Figure 6.18).

TIP: Show where the class files are backed up: an external hard disk, a flash drive or a CD, kept in another place.

TIP: Let pupils make eight cards of harmful software and match them with the damage they do (page 176 - 177).

WRITING SPACES

What Students Write

Read each note out or write it on the board - the students copy it into the space

Other pages

No.	Page	What the students write
1	2	**The four parts of this unit** ICT in entertainment; ethical and legal issues in ICT; precautions to protect ICT infrastructure; health and safety issues of ICT.

Chapter 1

No.	Page	What the students write
2	3	**3D films and holographic images** 3D films look real (3D spectacles). Holographic image processing shows a scene recorded in one place elsewhere.
3	4	**Games and simulation games** Digital games: computers and cell phones; 3D and 4D. Simulation games: an artificial natural setting, for training.

Chapter 2

No.	Page	What the students write
4	6	**Problems of using ICT** Legal issues, values, physical problems, harmful software, environmental, health related and social issues.
5	7	**Plagiarism** Displaying another's creative work as one's own. Avoid it: cite the owner, quote with "...", list the references.

Chapter 3

No.	Page	What the students write
6	10	**Physical security** UPS, hardware firewall, locked doors, CCTV, surge protector, and a safe place free from dirt, dust and moisture.
7	11	**Virus, worm and Trojan horse** A virus spreads within the computer; a worm spreads alone through e-mail and messages; a Trojan does not spread.

Chapter 4

No.	Page	What the students write
8	15	**Ergonomics** Greek: ergon (job, action) + nomos (law). The job must suit the comfort of the worker - not the other way round.
9	15	**Four health issues** Musculoskeletal problems, RSI (shoulder to fingers), CTS (fingers, wrist), CVS (eyes): wrong posture, long hours.

ANSWERS

Chapter 1

ICT in entertainment

A Match them

6 marks

1. 3D technology → C	**2.** Holographic image processing → F
3. Cartoon films → D	**4.** Digital audio materials → A
5. Digital games → E	**6.** Simulation games → B

1 mark each (textbook pages 168 - 170).

B Fill in the blanks

8 marks

1. film	**2.** 3D	**3.** spectacles	**4.** horror
5. mixing	**6.** discs	**7.** theatre	**8.** 4D

1 mark each. The extra words are **2D** and **cartoons**.

C True or false

6 marks

1. **True** - Film producers today make films of high quality with the use of ICT.
2. **False** - You do not need any spectacles to view a 3D film.
3. **True** - Cartoon films are popular among both adults and the young.
4. **False** - Digital games can be played only on computers, not on cell phones.
5. **True** - Simulation games are used for military training.
6. **False** - Holographic image processing is mostly used with comedy scenes.

D Choose the correct answer

5 marks

1. (b) three dimensional	**2.** (c) holographic image processing
3. (a) compact discs	**4.** (d) an artificial natural setting
5. (b) storing songs	

TEACHING TIPS

- Show a sample phishing e-mail made by you: a "bank" asks for an account number through a link.
- Prevention is better than cure: an updated virus guard, careful use of USBs, care with e-mail and links.
- Before Practical 2, check that the three websites of page 179 open in the lab.

ANSWERS

Chapter 2

Ethical and legal issues

A Which legal issue?

8 marks

1. Data theft	**2.** Unauthorized access	**3.** Intellectual property	**4.** Fraud
5. Data theft	**6.** Unauthorized access	**7.** Intellectual property	**8.** Fraud

1 mark for each line with the tick in this column (textbook page 171).

B Citing, quoting or referencing?

6 marks

1. Citing	**2.** Quoting	**3.** Referencing
4. Citing	**5.** Quoting	**6.** Referencing

1 mark each (textbook page 172).

C Choose the correct answer

5 marks

1. (a) Personally Identifiable Information	**2.** (c) the state
3. (b) plagiarism	**4.** (d) referencing
5. (c) fast e-mail	

D Answer these

4 marks

1. What are intellectual property rights? (2 marks)

An innovation or a completely new product is the property of the first person or organization to develop and release it (1 mark); it is their right to identify themselves as its legal or lawful owners (1 mark).

2. Write two better accepted ways of using information collected from the Internet. (2 marks)

Any two (1 mark each): citing - mention the rightful owner and his information; quoting - inverted commas ("...") around a borrowed section; referencing - list the resources at the end of the essay or article.

TEACHING TIP

- Ask pupils to sit as they sit at home in front of a computer. Discuss the posture with the picture of 6.13.

ANSWERS

Chapter 3

Protecting ICT infrastructure

A Physical or logical?

8 marks

1. A UPS: **Physical**	**2.** A password: **Logical**
3. A CCTV camera: **Physical**	**4.** A software firewall: **Logical**
5. A surge protector: **Physical**	**6.** A backup on an external hard disk: **Logical**
7. Closed and locked doors: **Physical**	**8.** A hardware firewall in a broadband router: **Physical**

1 mark each. In the textbook the **hardware** firewall is physical security and the **software** firewall is logical security.

B Match the harmful software

8 marks

1. Computer virus → D	**2.** Computer worm → G
3. Trojan horse → B	**4.** Adware → E
5. Bots → H	**6.** Browser hijacker → C
7. Phishing → F	**8.** Spam → A

1 mark each (the table of textbook pages 176 - 177).

C True or false

6 marks

1. **True** - A firewall protects a network from unauthorized access.
2. **False** - A software firewall needs to be active only once a week.
3. **False** - A Trojan horse spreads by itself, like a computer virus.
4. **True** - Adware displays unnecessary notices on the computer screen.
5. **True** - Phishing uses e-mail to collect information about bank accounts.
6. **False** - A backup is safest when it is kept right next to the computer.

D Choose the correct answer

5 marks

1. (c) a surge protector	**2.** (b) a password
3. (a) browser hijacker	**4.** (d) suspicious links
5. (b) ICTA	

TIP: Show the virus guards of page 178 and ask which one the school computers use.

E Word search

10 marks

P		A	V	I	R	U	S			L	
H		D		S					L		
I		W		W	P			A			
S		A		O		Y	W				
H	S	R		R		E	W				
I	T	E		M	R			A			
N	O			I					R		
G	B		F	T	R	O	J	A	N	E	
	H	I	J	A	C	K	E	R			
							M	A	P	S	

Word	Starts at	Direction
VIRUS	row 1, col 4	across
WORM	row 3, col 5	down
TROJAN	row 8, col 5	across
ADWARE	row 1, col 3	down
SPYWARE	row 2, col 5	slanting down
BOTS	row 8, col 2	up
HIJACKER	row 10, col 2	across
PHISHING	row 1, col 1	down
SPAM	row 11, col 11	across, backwards
FIREWALL	row 8, col 4	slanting up

1 mark for each word found and ticked.

ANSWERS

Chapter 4

Health issues and ergonomics

A Which health issue?

8 marks

1. Musculoskeletal	**2.** RSI	**3.** CTS	**4.** CVS
5. RSI	**6.** CTS	**7.** CVS	**8.** Musculoskeletal

1 mark each (textbook page 180).

B Where does it hurt?

4 marks

1. Computer Vision Syndrome (CVS)	**2.** Musculoskeletal problems
3. Repetitive Stress Injury (RSI)	**4.** Carpal Tunnel Syndrome (CTS)

1 mark each. Accept the short names CVS, RSI and CTS.

C Choose the correct answer

5 marks

1. (b) Greek	**2.** (c) four hours
3. (d) fingers	**4.** (a) dry eyes
5. (c) Repetitive Stress Injury	

MARKING SCHEME

Unit 6 Test

Total 40 marks

1 Choose the correct answer

8 marks (1 each)

1. (a) the 3D feature	**2.** (b) software
3. (c) hijacker	**4.** (d) 240 V
5. (a) spam	**6.** (b) "nomos"
7. (c) fast e-mail	**8.** (a) A and C only

2 Fill in the blanks

8 marks (1 each)

1. horror	**2.** 4D	**3.** numbers	**4.** patent
5. routers	**6.** incoming	**7.** Phishing	**8.** cyber

3 Name the pictures

8 marks (1 each)

1. 3D spectacles	**2.** Simulation game	**3.** Digital game	**4.** Digital audio
5. UPS	**6.** Surge protector	**7.** CCTV camera	**8.** Firewall

The extra words are **Backup** and **Password**. Also accept "3D glasses" (card 1), "driving simulator" (card 2) and "game controller" (card 3).

4 Answer these

16 marks

1. What are simulation games? Write one use of them. (2 marks)

Games based on an everyday activity in a natural setting created artificially (1 mark). Any one use (1 mark): training for sports activities, investigations, planning, military training, mercantile activities, miming.

2. What is plagiarism? Write two ways to avoid it. (3 marks)

Plagiarism is stealing another's creative work and displaying it as one's own (1 mark). Any two (1 mark each): citing (mention the rightful owner); quoting (inverted commas around a borrowed section); referencing (list the resources at the end).

3. Write two legal issues in the use of ICT and explain one of them. (3 marks)

Any two (1 mark each): data theft; unauthorized access into computer systems; intellectual property rights (stolen intellectual property); fraud. A correct explanation of one of them (1 mark), e.g. data theft - Personally Identifiable Information (PII) is stolen from a computer or another device.

4. What is a backup? Write two places where a backup can be kept. (2 marks)

A copy of the essential data and information of a computer, kept so that it is not lost when the system fails (1 mark). Any two (1 mark for both): an external hard disk, a CD, a DVD, a flash drive, a memory chip - securely, in another physical location.

5. How is a computer worm different from a computer virus? (2 marks)

A virus enters through a computer program as an executable file and spreads within the computer (1 mark). A worm acts and spreads alone, using e-mail attachments, false websites and instant messages (1 mark).

6. Write two ways to safeguard a computer from harmful software. (2 marks)

Any two (1 mark each): install a virus guard and keep it updated and active; check USBs with anti virus software; install authorized software; use a user account; access secure websites; check downloads with a virus guard; do not click suspicious links or open suspicious e-mails; use a firewall and e-mail filters.

7. Name two health issues of long computer use. (2 marks)

Any two (1 mark each): musculoskeletal problems; RSI - Repetitive Stress Injury; CTS - Carpal Tunnel Syndrome; CVS - Computer Vision Syndrome.

(i) WHAT STUDENTS USUALLY GET WRONG

- They call the hardware firewall logical security. In the textbook it is **physical** security; the software firewall is logical security.
- They mix up citing (naming the owner) and referencing (the list of sources at the **end**).
- They write that a Trojan horse spreads by itself. Viruses and worms spread; the Trojan horse **does not**.
- They mix up phishing (an e-mail and a link that steal account details) and spam (unauthorized e-mail, mostly advertisements).
- They mix up the UPS (a power failure: battery power) and the surge protector (a supply above 240 V).
- They mix up RSI (shoulder to fingers) and CTS (numbness in the fingers, pressure on the wrist).
- They spell "Carpel". The correct spelling is **Carpal** Tunnel Syndrome (the textbook prints "Carpel").

(★) TEACHING TIPS

- Remind pupils that health issues arise after four hours of continuous use. A short break helps.
- Read each writing-space note slowly or write it on the board, then check a few books each lesson.
- O/L Paper I has MCQs on these topics: let pupils do the MCQs of each chapter with a time limit.

HOMEWORK AND EXTRA PRACTICE

- List the films, music and games your family enjoys. Which ICT facility of 6.1 - 6.2 does each one use?
- Write five sentences on a topic of this unit with one quote, one citation and a reference list.
- Check a computer at home: is the virus guard updated? Is the firewall on? Is there a backup?
- Fast learners: make a poster about the four health issues of 6.13 and how to avoid them.

TEXTBOOK NOTES - THE WORKBOOK CORRECTS THESE

- Page 176: the harmful software table is misprinted - "Spyware" has no description, and "Trojan Horse" and "Adware" are printed in the damage column. The workbook gives them their own rows.
- Page 178: "Avast Antivirus AVG Antivirus" is two virus guards on one line (12 virus guards in all).
- Page 174: "The surge protects computers" means the surge **protector**. Page 173: "imporve" = improve.
- Page 171: "Intelectual property usualy reletes to creation in mind" and "photoes". Page 180: "Carpel".
- Page 170: "miming" among the uses of simulation games may be a misprint of "mining". The workbook keeps the textbook word.

TEACHING TIPS

- In O/L Paper II, pupils should use the textbook words: data theft, plagiarism, UPS, firewall, ergonomics.
- Mark the unit test together the next day. Pupils learn most from the questions they got wrong.
- Ask pupils for one ICT problem they met this month and put it under one of the issues of Figure 6.8.
- Use the word search as quick revision of the harmful software at the end of Chapter 3.
- Ask which entertainment at home uses ICT: CDs, home theatre systems, games on cell phones, cartoon films.
- Use a new mobile app made by a Sri Lankan student as the example of intellectual property: who owns it?
- Before the malware lesson, scan a USB flash drive with the virus guard in front of the class.
- Let pupils count the hours they spend at a computer or a phone in one day and compare them with page 179.